

The developer's cheatsheet

Docker, *end to end.*

Every command you actually need —
build, run, compose, ship.

`docker build` → `docker push`



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01 Core Docker CLI

The commands you'll type every day

docker build image from a Dockerfile

docker run container from an image

docker ps list containers (-a for all)

```
terminal

$ docker build -t myapp:latest .
$ docker run -d -p 3000:3000 myapp
$ docker ps -a
```



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02 Dockerfile Essentials

The three directives that matter most

FROM sets the base image

COPY copies files in

CMD the default command

```
Dockerfile

FROM node:18-alpine
COPY . /app
CMD ["npm", "start"]
```



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03 Core Concepts

Three words you'll say all day

Image a snapshot of your app + deps

Container a running instance of an image

Volume persists data when containers die

 terminal

```
$ docker run -v /data:/app/data myapp
```

Use volumes so you never lose data when a container is stopped or removed.



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04 Docker Compose

Run multi-container apps from one file

docker compose up -d starts everything, detached

```
docker-compose.yml

services:
  app:
    build: .
    ports:
      - '3000:3000'
  mongo:
    image: mongo
    volumes:
      - mongo-data:/data/db
volumes:
  mongo-data:
```



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05 Networking & Ports

Connect containers and expose them

Port binding maps container ports to the host

Bridge the default network

--network join containers on one net

 terminal

```
$ docker run -p 8080:80 nginx
```

```
$ docker network create mynet
```

```
$ docker run --network=mynet --name=db mongo
```



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06 Optimizations

Smaller images, faster builds

Multi-stage build heavy, ship light

.dockerignore exclude files from the build

Dockerfile

```
FROM node:18 AS builder
WORKDIR /app
COPY . .
RUN npm install && npm run build

FROM nginx:alpine
COPY --from=builder /app/dist /usr/share/nginx/html
```

.dockerignore

```
node_modules
*.log
dist
```



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07 Debugging & Logs

When a container misbehaves

docker logs see container output

docker exec run commands inside

docker inspect full container details

 terminal

```
$ docker logs <container-id>
```

```
$ docker exec -it <container-id> /bin/sh
```

```
$ docker inspect <container-id>
```



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08 Hosting & CI/CD

Tag, push, and automate

Docker Hub the public image registry

```
terminal

$ docker tag myapp username/myapp:1.0
$ docker push username/myapp:1.0
```

Wire it into CI with GitHub Actions:

```
.github/workflows/ci.yml

jobs:
  docker:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v2
      - run: |
          docker build -t myapp .
          docker push username/myapp
```



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Docker trial and error.

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